

Eliminating Reversals from Cubical Type Theories

Evan Cavallo

*University of Gothenburg and
Chalmers University of Technology*

Christian Sattler

*Chalmers University of Technology
and University of Gothenburg*

Homotopical type theories

- Recall propositions as types methodology
 - no distinction between **proofs** and **constructions**:

M is a proof of the proposition A $\xrightarrow[\text{becomes}]{\sim}$ $M : A$

- To fit **equality propositions** seamlessly in this picture, we'd like **equality proofs** to be able to carry **data**
 - e.g., if we want to include **quotients** in our type theory
 - $[a]$ and $[b]$ the same in A/R $\iff$ there is a $w : R(a, b)$

- Homotopical type theories address these thorny issues

Cubical type theory

▣ Instead of equalities, **paths**: functions from an **interval** “type”



▣ Talk about paths using **path types**:

$$\begin{aligned} i : \mathbb{I} \vdash p(i) : A &\rightsquigarrow \lambda i. p(i) : \text{Path}_A(a_0, a_1) \\ p(0) &= a_0 : A \\ p(1) &= a_1 : A \end{aligned}$$

▣ Quotients, for example, get a **path constructor**:

$$\begin{aligned} w : R(a, b) &\rightsquigarrow i : \mathbb{I} \vdash \text{rel}(w, i) : A/R \\ \text{rel}(w, 0) &= [a] : A/R \\ \text{rel}(w, 1) &= [b] : A/R \end{aligned}$$

▣ Think of types as looking like **(topological) spaces**

Cubical type theory: coercion

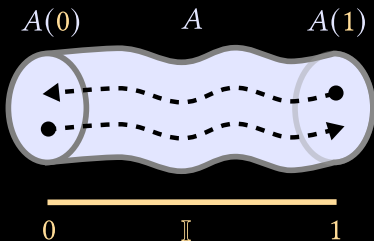
⊞ Operator called **coercion** makes paths act like equalities

▣ All elements of $\mathbb{I}$ are “equal”

▣ so: for $i : \mathbb{I} \vdash A$ type, the types $A(r)$ and $A(s)$ should “look the same” for all $r, s : \mathbb{I}$

$$\frac{i : \mathbb{I} \vdash A(i) \text{ type} \quad r, s : \mathbb{I}}{\text{coe}_A^{r \rightarrow s} : A(r) \rightarrow A(s)}$$

$$\text{coe}_A^{r \rightarrow r} = \text{id}_{A(r)}$$



▣ Hard part of designing a cubical type theory is **making coe compute**

Cubical type theories, plural

▣ Comes in many varieties [CCHM15, AFH18, ABCHFL21, ...]

▣ We only **need**

$$\overline{0 : \mathbb{I}}$$

$$\overline{1 : \mathbb{I}}$$

but it could be **convenient** to include more features of (say) the topological interval, e.g.:

$$\frac{r, s : \mathbb{I}}{r \vee s : \mathbb{I}}$$

$$r \vee 0 = r$$

$$r \vee 1 = 1$$

...

▣ Tension between **convenience** and **well-behaved semantics**

▣ **Semantic goal**: use cubical type theory to reason about **the homotopy theory of spaces**

Cubical type theories with reversal

▣ Today we look at **reversal**:

$$\frac{r : \mathbb{I}}{\neg r : \mathbb{I}} \quad \neg 0 = 1 \quad \neg \neg r = r$$



▣ Practically useful – makes it easy to **invert** paths

$$\begin{aligned} \text{sym} &: \text{Path}_A(a_0, a_1) \rightarrow \text{Path}_A(a_1, a_0) \\ \text{sym}(p) &:= \lambda i. p(\neg i) \end{aligned}$$

▣ satisfies $\text{sym}(\text{sym}(p)) = p$ up to **strict equality**

▣ cut down on coherences in **synthetic homotopy theory**

▣ Concretely: in Agda, Mikan proof assistants along with $\wedge$, $\vee$

▣ Unlike $\vee$ and $\wedge$, no model in spaces was known

Results

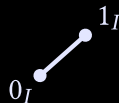
- ▣ We show that a reversal $\neg$ is mostly harmless in two ways
 - ▣ Procedure for **upgrading semantic models** of cubical type theory to add $\neg$
 - ▣ Get a semantics of $\neg$ in spaces
 - ▣ For a weaker “opaque” cubical type theory, prove a **conservativity** result for extension by $\neg$
 - ▣ Hope: stepping stone to result for full CTT
- ▣ Everything builds on one geometric trick
 - ▣ **Simplest non-trivial case** – trial run towards general results?

I. One Weird Trick

Twist constructions

- ▣ Suppose I is a **bipointed set**, a set with two chosen elements

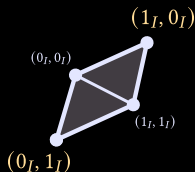
$$0_I \in I \qquad 1_I \in I$$



- ▣ Have a bipointed set I^{op} with same underlying set but $0_{I^{\text{op}}} = 1_I$ and $1_{I^{\text{op}}} = 0_I$

- ▣ Take $\mathsf{TI} := I \times I^{\text{op}}$, so:

$$0_{\mathsf{TI}} := (0_I, 1_I) \qquad 1_{\mathsf{TI}} := (1_I, 0_I)$$



- ▣ TI has a “reversal” $\neg: \mathsf{TI} \cong (\mathsf{TI})^{\text{op}}$ that permutes coordinates:

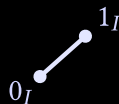
$$\neg(i, j) := (j, i)$$



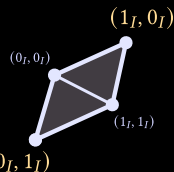
Twist constructions

▣ Suppose I is a distributive lattice:

$$\begin{array}{lll} 0_I \in I & \vee_I: I \rightarrow I \rightarrow I & + \text{ lots of} \\ 1_I \in I & \wedge_I: I \rightarrow I \rightarrow I & \text{equations} \end{array}$$



▣ Have a dual distributive lattice I^{op} with
 $0_{I^{\text{op}}} = 1_I$ and $1_{I^{\text{op}}} = 0_I$ and $\wedge_{I^{\text{op}}} = \vee_I$ and $\vee_{I^{\text{op}}} = \wedge_I$



▣ Take $TI := I \times I^{\text{op}}$, so, e.g.:

$$(i_0, i_1) \vee_{TI} (j_0, j_1) := (i_0 \vee_I j_0, i_1 \wedge_I j_1)$$

▣ This distributive lattice is a De Morgan algebra

$$\neg(i, j) := (j, i)$$



Twist constructions

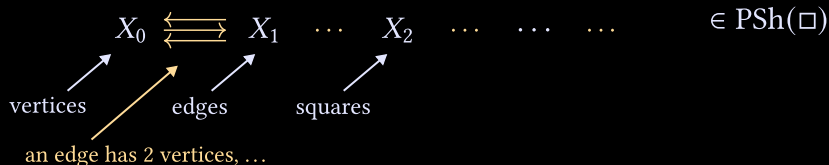
- ⊞ Any “self-dual” algebraic theory will support a **T** construction
- ⊞ Such constructions appear in e.g. order theory, logic literature
- ⊞ We adopted the name “**twist construction**” which originates with Kracht (1998)
- ⊞ From an interval w/o reversal, we can get an interval w/ reversal
 - ▣ ... in cubical set semantics
 - ▣ ... inside type theory

II. Scaffolding: Semantics

Cubical set semantics

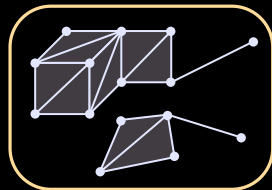
▣ Since Bezem, Coquand, and Huber (2013), we build semantics interpreting types as **cubical sets**

▣ Cubical set X = “combinatorial” description of a space as made by attaching n -dimensional cubes to each other



▣ “cubical set” can be defined in different ways by adding more maps between X_i ’s

▣ e.g., $\neg_i: X_n \rightarrow X_n$
reversing the i th axis of an n -cube

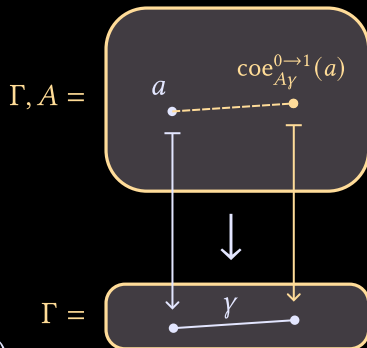


Cubical set semantics of types

- ▣ A **dependent type** $\Gamma \vdash A$ type is interpreted as a cubical set **over** a cubical set
- ▣ ...that is a **Kan fibration**
- ▣ In particular, a Kan fibration must be able to interpret **coercion**

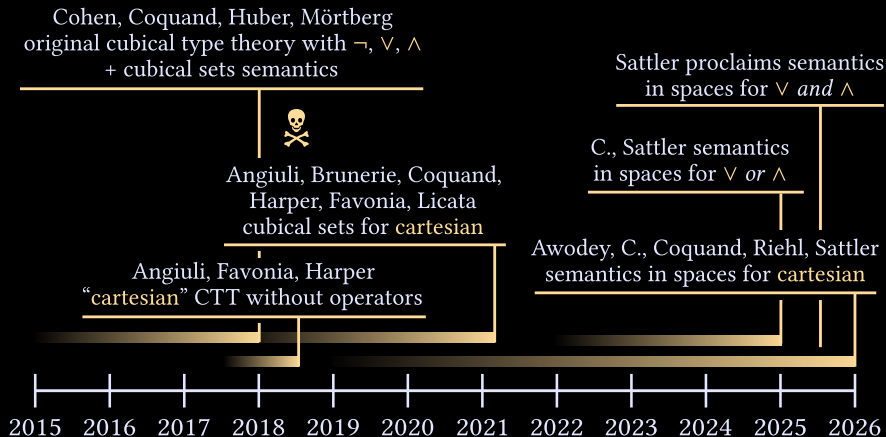
$$\frac{i : \mathbb{I} \vdash A(i) \text{ type} \quad r, s : \mathbb{I}}{\text{coe}_A^{r \rightarrow s} : A(r) \rightarrow A(s)}$$

(and more that we forget about today)



- ▣ Variety of cubical set + class of fibrations determines a **homotopy theory** we can compare to the **homotopy theory of spaces**

Some history



Does any CTT have a semantics with the *homotopy theory of spaces*?

☠ Sattler: known semantics of CTTs have pathological homotopy theories

A good model to twist

▣ Angiuli, Brunerie, Coquand, Harper, Favonia, and Licata proved something more general than

Thm (ABCHFL '21): Let $\mathcal{C}$ be a finite product category containing an interval $1 \rightrightarrows I$. There is a semantics in $\mathbf{PSh}(\mathcal{C})$ for CTT with interval operators $f: I^n \rightarrow I$ in $\mathcal{C}$.

▣ Usually, $\mathcal{C}$ is a cube category – every object is a power I^n

▣ Types are interpreted as Kan fibrations, as defined with I

▣ Often this semantics does **not** have the homotopy theory of spaces (💀), but

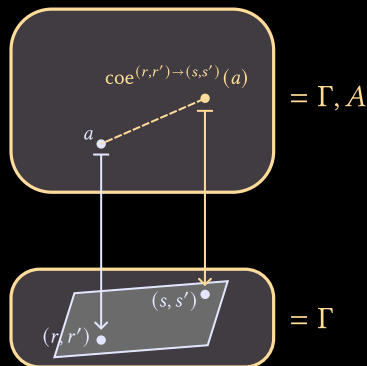
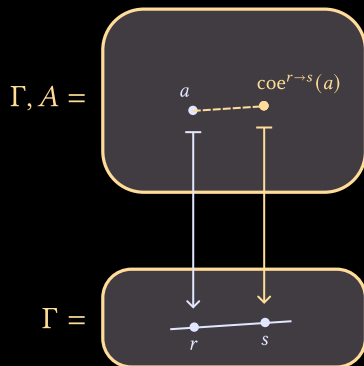
Thm (C.–Sattler '25): There is a $(\mathcal{C}, 1 \rightrightarrows I)$ (cartesian cubes + $\vee$) whose ABCHFL semantics has the homotopy theory of spaces

Let's twist it

⊞ If $1 \xrightarrow[1]{0} I \in \mathcal{C}$ is an ABCHFL input, so is $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

Thm: ABCHFL semantics for $1 \xrightarrow[1]{0} I \in \mathcal{C}$ and $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

have the same semantic types ($\implies$ same homotopy theory)

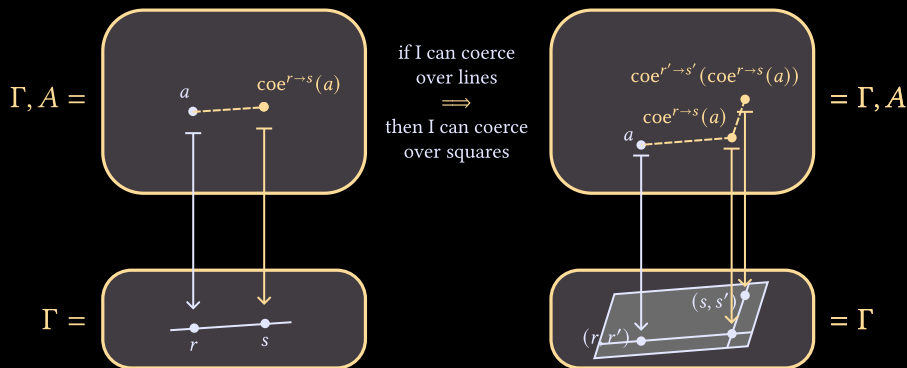


Let's twist it

⊞ If $1 \xRightarrow[1]{0} I \in \mathcal{C}$ is an ABCHFL input, so is $1 \xRightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

Thm: ABCHFL semantics for $1 \xRightarrow[1]{0} I \in \mathcal{C}$ and $1 \xRightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

have the same semantic types ($\Rightarrow$ same homotopy theory)

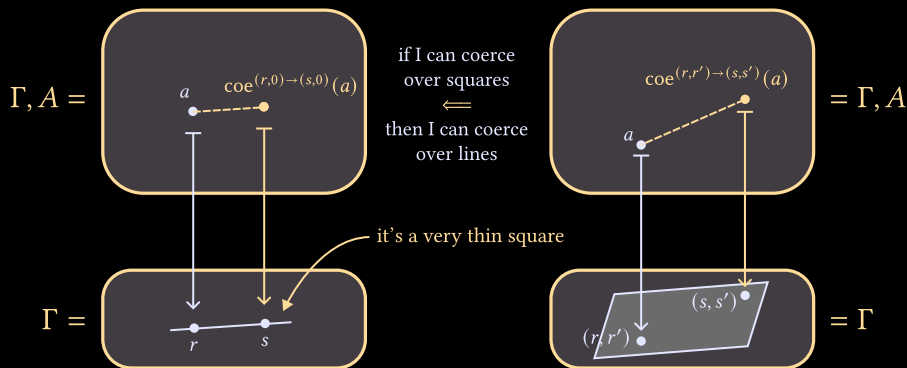


Let's twist it

⊞ If $1 \xrightarrow[1]{0} I \in \mathcal{C}$ is an ABCHFL input, so is $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

Thm: ABCHFL semantics for $1 \xrightarrow[1]{0} I \in \mathcal{C}$ and $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

have the same semantic types ($\implies$ same homotopy theory)

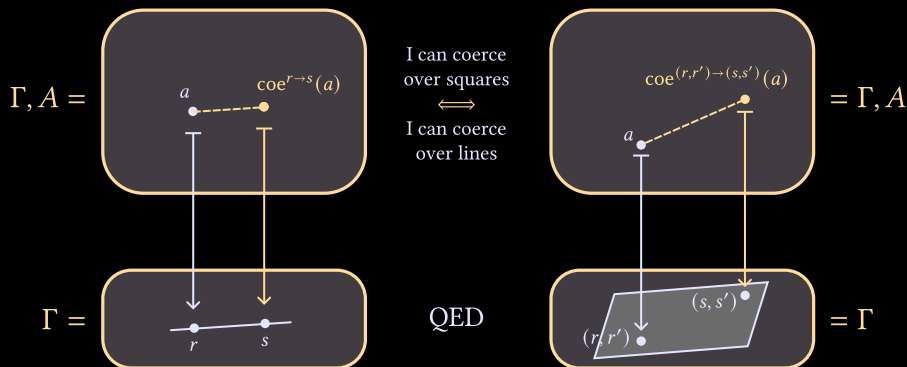


Let's twist it

⊞ If $1 \xrightarrow[1]{0} I \in \mathcal{C}$ is an ABCHFL input, so is $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

Thm: ABCHFL semantics for $1 \xrightarrow[1]{0} I \in \mathcal{C}$ and $1 \xrightarrow[(1,0)]{(0,1)} I \times I \in \mathcal{C}$

have the same semantic types ($\implies$ same homotopy theory)



Putting it together

⊞ A semantics of cubical type theory with $\neg$ in spaces!

Thm: There is an ABCHFL input $(C, 1 \rightrightarrows J)$ with reversal whose ABCHFL semantics has the homotopy theory of spaces.

Pf: Take $C = \text{cartesian cubes} + \vee$ and $J = I \times I^{\text{op}}$.

⊞ Combining $\wedge$, $\vee$, and $\neg$ (Cubical Agda / Mikan)?

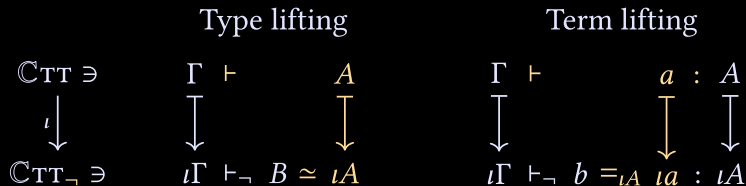
▣ Open whether ABCHFL semantics for $C = \text{cartesian} + \wedge + \vee$ has homotopy theory of spaces

▣ But twist should also apply to Sattler's modified semantics of $\wedge$ and $\vee$, which does

III. Scaffolding: Type Theories

Conservativity

- ▣ I'd rather not think about specific model constructions
- ▣ Better: **twist translation** T from cubical type theories with $\neg$ to cubical type theories without it
- ▣ Even better: use T to prove a **conservativity** property
 - show $\iota : \mathbb{C}_{\mathsf{TT}} \rightarrow \mathbb{C}_{\mathsf{TT}\neg}$ is a **weak equivalence**



- ▣ Can almost do this, but...

Conservativity for “opaque” type theory

- Our translation T does not preserve strict equations used to evaluate coe , e.g.

$$\text{coe}_{A \times B}^{r \rightarrow s}(\langle a, b \rangle) = \langle \text{coe}_A^{r \rightarrow s}(a), \text{coe}_B^{r \rightarrow s}(b) \rangle$$

- So we define it on “opaque” cubical type theories without these equations:

Thm: For $\mathbb{C}TT$ an opaque cubical type theory with self-dual interval structure (e.g., $\wedge$ and $\vee$), $\mathbb{C}TT_{\neg}$ is conservative over $\mathbb{C}TT$.

- Significant sacrifice, but some reason to hope ordinary $\mathbb{C}TT$ is also conservative over “opaque” $\mathbb{C}TT$
- Coquand, Huber, and Sattler (2022)

Conservativity for “opaque” type theory

Thm: For $\mathbb{C}TT$ an opaque cubical type theory with self-dual interval structure (e.g., $\wedge$ and $\vee$), $\mathbb{C}TT_{\neg}$ is conservative over $\mathbb{C}TT$.

⌘ Get a lot of help from general tools

- ▣ Uemura’s (2021, 2023) **second-order generalized algebraic theories (SOGATs)** – a logical framework – help us define and manipulate “cubical type theories”
- ▣ Kapulkin–Lumsdaine’s (2018) theory of **weak equivalences** help us turn a translation into a conservativity result

⌘ Do some steps in some generality

Looking at the future

- ▣ We had the opportunity to work out the scaffolding around an extremely simple translation $\mathbb{I} \mapsto \mathbb{I} \times \mathbb{I}^{\text{op}}$
- ▣ Some potentially fruitful ideas:
 - ▣ Let's look at theories, not just models
 - ▣ Try to separate conservativity of “full” over “opaque” CTT from comparisons between interval theories
- ▣ Conjecture: **any** extension of the interval theory of a CTT will be a conservative extension

THANKS